

Nils Lukas

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Research Interests

Design secure and private Machine Learning systems in the presence of untrustworthy

1. **Providers:** Confidential computing via Homomorphic Encryption & Secret Sharing.
2. **Data:** Mitigate data poisoning during training & prompt injection during inference.
3. **Models:** Protect training data privacy through PII scrubbing & differential privacy.
4. **Users:** Control misuse by detecting generated (mis)information with watermarking.

Education

University of Waterloo, Canada 2019 – 02/2024
Ph.D. in Computer Science

- Advisor: [Florian Kerschbaum](#)
- Thesis: Analyzing Threats of Large-Scale Machine Learning Systems
- Thesis Awards: [Top Mathematics Doctoral Prize](#) & [Alumni Gold Medal](#)

RWTH-Aachen, Germany
M.Sc. in Computer Science (*w/Distinction*) 2016 – 2018
B.Sc. in Computer Science 10/2012 – 2016

Honors & Awards

[Amazon Research Award](#) [100,000 USD] 2025
First Place at the NeurIPS'24 Watermarking Competition [4,400 USD] 2024
First Place at DGE Elite Hackathon, GITEX'24 [10,900 USD] 2024
Top Mathematics Doctoral Thesis, University of Waterloo [1,080 USD] 2024
Alumni Gold Medal, One PhD Award Yearly, University of Waterloo 2024
Best Poster Award, Sponsored by David R. Cheriton [220 USD] 2023
Distinguished Contribution Award, Microsoft MLADS conference 2023
David R. Cheriton Scholarship, University of Waterloo [14,400 USD] 2022, 2023
Outstanding Reviewer, ICML'22 2022
Best Poster Award, Sponsored by Rogers [720 USD] 2019

Research Grants

Awarded **\$1,458,000**
[United AI-Saqer Group] MedBoard: Privacy-preserving Multi-Agent AI for Healthcare **\$136,000** · 2026
PI: Nils Lukas · Co-PIs: [Praneeth Vepakomma](#), [Preslav Nakov](#)
[IAAI Research Program] IAAI Knowledge Pipeline **\$500,000** · 2026
PI: [Salem Lahlou](#) · Co-PIs: Nils Lukas, [Martin Takáč](#)
[Amazon Research Awards – Special Call] AdvSim2Real: Robustifying Agents in Simulated Adversarial Environments **\$100,000** · 2025
PI: Nils Lukas · Co-PI: [Praneeth Vepakomma](#)
[Etihad Airways] Conversational Booking Agents **\$450,000** · 2025
PI: Nils Lukas · Co-PIs: [Salem Lahlou](#), [Alham Fikri Aji](#), [Martin Takáč](#), [Mingming Gong](#)
[United AI-Saqer Group] Privacy-preserving Brain–Computer Interfaces **\$136,000** · 2025
PI: [Abdulrahman Mahmoud](#) · Co-PIs: Nils Lukas, [Elizabeth Churchill](#)
[TII Funding] GFlowNets for Fuzzing of Agentic Applications **\$136,000** · 2025
PIs: [Salem Lahlou](#) & Nils Lukas

Conference Publications

23
[ECCV'26] [SPQR: A Multi-Dimensional Benchmark for Safety Alignment under Benign Model Adaptation](#)
Mohammed Talha Alam, Nada Saadi, Fahad Shamshad, Nils Lukas, Karthik Nandakumar, Fakhri Karray, Samuele Poppi. European Conference on Computer Vision, 2026.

22	RAVEN: Erasing Invisible Watermarks via Novel View Synthesis Fahad Shamshad, Nils Lukas , Karthik Nandakumar. IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2026.
[CVPR'26] 🏆 Oral (top 0.8%)	
21	Collaborative Threshold Watermarking Tameem Bakr, Anish Ambreth, Nils Lukas . The Forty-Third International Conference on Machine Learning, 2026.
[ICML'26]	
20	CoRe: Collaborative Reasoning via Cross Teaching Kshitij Mishra, Mirat Aubakirov, Martin Takáč, Nils Lukas , Salem Lahlou. The Forty-Third International Conference on Machine Learning, 2026.
[ICML'26]	
19	Forest Before Trees: Latent Superposition for Efficient Visual Reasoning Yubo Wang, Juntian Zhang, Yichen Wu, Yankai Lin, Nils Lukas , Yuhan Liu. The 64th Annual Meeting of the Association for Computational Linguistics, 2026.
[ACL'26] Main Conference	
18	Detecting AI-Generated Video: A Vision-Language Dual-View Survey Dylan Xinming Hou, Juntian Zhang, Xu Gu, Yichen Wu, Nils Lukas , Gus Xia, Xiuying Chen, Yuhan Liu. Findings of the Association for Computational Linguistics, 2026.
[ACL'26] Findings	
17	DP-Fusion: Token-Level Differentially Private Inference for Large Language Models Rushil Thareja, Preslav Nakov, Praneeth Vepakomma , Nils Lukas . The Fourteenth International Conference on Learning Representations, 2026.
[ICLR'26]	
16	SD-E²: Semantic Exploration for Reasoning Under Token Budgets Kshitij Mishra, Nils Lukas , Salem Lahlou. Findings of the European Chapter of the Association for Computational Linguistics, 2026.
[EACL'26] Findings	
15	Mask Image Watermarking Runyi Hu, Jie Zhang, Shiqian Zhao, Nils Lukas , Jiwei Li, Qing Guo, Han Qiu, Tianwei Zhang. The Thirty-Ninth Annual Conference on Neural Information Processing Systems, 2025.
[NeurIPS'25] AR: 24.5% (5,290/21,575)	
14	SPIRIT: Patching Speech Language Models against Jailbreak Attacks Amir Djanibekov, Nurdaulet Mukhituly, Kentaro Inui, Hanan Aldarmaki, Nils Lukas . Empirical Methods in Natural Language Processing (Main Conference), 2025.
[EMNLP'25] AR: 22.2% (1,811/8,172)	
13	Optimizing Adaptive Attacks against Content Watermarks for Language Models Abdulrahman Diaa, Toluwani Aremu, Nils Lukas . The Forty-Second International Conference on Machine Learning, 2025.
[ICML'25] AR: 26.9% (3,260/12,107) ★ Spotlight (Top 2.6%)	
12	Cowpox: Towards the Immunity of VLM-based Multi-Agent Systems Yutong Wu, Jie Zhang, Yiming Li, Chao Zhang, Qing Guo, Han Qiu, Nils Lukas , Tianwei Zhang. The Forty-Second International Conference on Machine Learning, 2025.
[ICML'25] AR: 26.9% (3,260/12,107)	
11	PEPSI: Practically Efficient Private Set Intersection in the Unbalanced Setting Rasoul Mahdavi, Nils Lukas , Faezeh Ebrahimianghazani, Thomas Humphries, Bailey Kacsmar, John Premkumar, Xinda Li, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
[USENIX'24] AR: 19.1% (417/2,176)	

10	Fast and Private Inference of Deep Neural Networks by Co-designing Activation Functions
[USENIX'24]	Abdulrahman Diaa, Lucas Fenaux, Thomas Humphries, Marian Dietz, Faezeh Ebrahimiaghazani, Bailey Kacsmar, Xinda Li, Nils Lukas , Rasoul Akhavan Mahdavi, Simon Oya, Ehsan Amjadian, Florian Kerschbaum. In the 33rd USENIX Security Symposium, 2024.
AR: 19.1% (417/2,176)	
9	Leveraging Optimization for Adaptive Attacks on Image Watermarks
[ICLR'24]	Nils Lukas , Abdulrahman Diaa, Lucas Fenaux, Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
AR: 30.8% (2,250/7,262)	
8	Universal Backdoor Attacks
[ICLR'24]	Benjamin Schneider, Nils Lukas , Florian Kerschbaum. In the Twelfth International Conference on Learning Representations, 2024.
AR: 30.8% (2,250/7,262)	
📰 Media Coverage	
7	PTW: Pivotal Tuning Watermarking for Pre-Trained Image Generators
[USENIX'23]	Nils Lukas and Florian Kerschbaum. In the 32nd USENIX Security Symposium, 2023.
AR: 29.2% (422/1,444)	
6	Analyzing Leakage of Personally Identifiable Information in Language Models
[S&P'23]	Nils Lukas , Ahmed Salem, Robert Sim, Shruti Tople, Lukas Wutschitz, Santiago Zanella-Béguelin. In the 44th IEEE Symposium on Security and Privacy, 2023.
AR: 17.0% (195/1,147)	
★ Distinguished Contribution Award at Microsoft MLADS	
5	SoK: How Robust is Image Classification Deep Neural Network Watermarking?
[S&P'22]	Nils Lukas , Edward Jiang, Xinda Li, Florian Kerschbaum. In the 43rd IEEE Symposium on Security and Privacy, 2022.
AR: 14.5% (147/1,012)	
4	Deep Neural Network Fingerprinting by Conferrable Adversarial Examples
[ICLR'21]	Nils Lukas , Yuxuan Zhang, Florian Kerschbaum. The Ninth International Conference on Learning Representations, 2021.
AR: 28.7% (860/2,997)	
★ Spotlight (Top 5%)	
3	On the Robustness of Backdoor-based Watermarking in Deep Neural Networks
[IH&MMSEC'21]	Masoumeh Shafieinejad, Nils Lukas , Jiaqi Wang, Xinda Li, Florian Kerschbaum. Proceedings of the 2021 ACM Workshop on Information Hiding and Multimedia Security, 2021.
AR: 40.3% (128/318)	
2	Practical Over-Threshold Multi-Party Private Set Intersection
[ACSAC'20]	Rasoul Mahdavi, Thomas Humphries, Bailey Kacsmar, Simeon Krastnikov, Nils Lukas , John Premkumar, Masoumeh Shafieinejad, Simon Oya, Florian Kerschbaum, Erik-Oliver Blass. Annual Computer Security Applications Conference (ACSAC), 2020.
AR: 20.9% (104/497)	
1	Differentially Private Two-Party Set Operations
[EuroS&P'20]	Bailey Kacsmar, Basit Khurram, Nils Lukas , Alexander Norton, Masoumeh Shafieinejad, Zhiwei Shang, Yaser Baseri, Maryam Sepehri, Simon Oya, Florian Kerschbaum. IEEE European Symposium on Security and Privacy (EuroS&P), 2020.
AR: 20.9% (39/187)	
Journal Publications	
2	Privacy-Preserving Machine Learning [Cryptography]
[IEEE S&P'23]	Florian Kerschbaum, Nils Lukas . IEEE Security & Privacy, Volume 21, Issue 6, 2023.

	¹ [AIP'18]	SunFlower: A new Solar Tower Simulation Method for use in Field Layout Optimization Pascal Richter, Gregor Heiming, Nils Lukas , Martin Frank. AIP Conference Proceedings, Volume 2033, Issue 1, 2018.	
Workshop Papers	³ [GenAI4Health]	Sanitizing Medical Documents with Differential Privacy using Large Language Models Rushil Thareja, Gautam Gupta, Preslav Nakov, Praneeth Vepakomma, Nils Lukas . 2025.	
	² [WMARK'25]	First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge Fahad Shamshad, Tameem Bakr, Yahia Salaheldin Shaaban, Noor Hazim Hussein, Karthik Nandakumar and Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.	
	¹ [WMARK'25] ★ Oral Presentation	Optimizing Adaptive Attacks against Content Watermarks for Language Models Abdulrahman Diaa, Toluwani Aremu, Nils Lukas . The 1st Workshop on GenAI Watermarking, 2025.	
Working Papers		Mitigating Watermark Forgery in Generative Models via Randomized Key Selection , Submitted. Toluwani Aremu, Noor Hazim Hussein, Munachiso S Nwadike, Samuele Poppi, Jie Zhang, Karthik Nandakumar, Neil Zhenqiang Gong, Nils Lukas . 2025. Robust and Calibrated Detection of Authentic Multimedia Content , Submitted. Sarim Hashmi, Abdelrahman Elsayed, Mohammed Talha Alam, Samuele Poppi, Nils Lukas . 2025.	
Research Talks		Adaptively Robust and Forgery-Resistant Watermarking • Meta (FAIR), hosted by Hady Elsahar	2025
		Optimizing Adaptive Attacks against Content Watermarks • DeepMind, hosted by David Stutz	2024
		• University of California, Berkeley, hosted by Dawn Song	2024
		Analyzing Leakage of Personal Information in Language Models • Microsoft M365, hosted by Robert Sim	2024
		• Meta, hosted by Will Bullock	2023
		• MongoDB, hosted by Marilyn George and Archita Agarwal	2023
		How Reliable is Watermarking for Image Generators? • Google, hosted by Somesh Jha	2023
		• University of California, Berkeley, hosted by Dawn Song	2023
Keynotes		Aviation Future Week , hosted by Emirates, Dubai	2024
		Cyber Energy Leadership Forum , Abu Dhabi	2024
Work Experience		Assistant Professor , MBZUAI, Abu Dhabi, UAE	since 08/2024
		Research Intern , Royal Bank of Canada, Borealis AI, Toronto	2024
		• Vertical Federated Learning, hosted by Kevin Wilson	
		Research Intern , Microsoft Research, Cambridge, UK	2022
		• Privacy for Language Models, hosted by Shruti Tople & Lukas Wutschitz	
		Research Assistant , RWTH-Aachen, Aachen	2014 – 2018
		Student Researcher , DSA Daten- und Systemtechnik GmbH, Aachen	2016
		Software Engineer Intern , A.R. Bayer DSP Systeme GmbH, Düsseldorf	2012
Teaching		Instructor , MBZUAI, UAE	

Service	• ML8509: Collaborative Machine Learning (w/ S. Horvath)	2026
	• ML8502: Machine Learning Security (14 weeks)	2025
	• ML807: Federated Learning (7 weeks)	2025
	• ML818: Emerging Topics in Trustworthy Machine Learning (4 weeks)	2024
	Teaching Assistant, University of Waterloo, Canada	
	• CS458/658: Computer Security and Privacy	2020, 2021
	• CS246 – Object Oriented Programming	2021
	Co-Instructor, RWTH-Aachen, Germany	
	• Course: Data-driven Medicine	2018
	Area Chair	
	• International Conference on Learning Representations (ICLR)	2026
	• International Conference on Machine Learning (ICML)	2026
	Program Committee	
	• ACM Conference on Computer and Communications Security (CCS)	2025
	• IEEE Symposium on Security and Privacy (IEEE S&P)	2025, 2026
	• Recent Advances in Intrusion Detection (RAID)	2024
	Artifact Evaluation Committee	
	• The ACM Conference on Computer and Communications Security (CCS)	2023, 2024
	Reviewer	
	• NETYS	2025
	• ACM TheWebConf (WWW)	2025
	• International Conference on Learning Representations (ICLR)	2024, 2025
	• International World Wide Web Conference (TheWebConf)	2024
	• Recent Advances in Intrusion Detection (RAID)	2023
	• Neural Information Processing Systems (NeurIPS)	2022, 2023
	• International Conference on Machine Learning (ICML)	2022, 2025
	• The Conference on Information and Knowledge Management (CIKM)	2020
PostDocs, Students	Other	
	• Sub-Reviewer, Proceedings on Privacy Enhancing Technologies (PETS)	2021–2023
	• Session Chair, IEEE Symposium on Security and Privacy (S&P)	2023
	• Organization, Workshop on Semantic Web Solutions for Large-Scale Biomedical Data Analytics (SeWeBMeDA)	2018
	• Chair for the invited faculty talk program, International Symposium on Trustworthy Foundation Models at MBZUAI	2025
	• Faculty Search Committee, Machine Learning Department at MBZUAI	2025
	• Admission's Committee, MBZUAI Machine Learning Department	2025
	Student Board Member, Cybersecurity and Privacy Institute	2022–2024
	School Advisory Committee on Appointments Liaison, CrySP Lab	2022
	Postdoctoral Researchers	
	• Samuele Poppi <i>[primary]</i>	since 2025
	• Kshitij Mishra <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)	since 2025
	• Dushyant Chauhan <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)	since 2025
	• Yuhan Liu <i>[secondary]</i> (w/ Martin Takáč & Salem Lahlou)	since 2025
	PhD Students	
	• Fabian Zhang (ML)	since 2025
	• Tair Djanibekov (NLP)	since 2025
	• Rushil Thareja (NLP) <i>[primary]</i> (w/ Praneeth Vepakomma)	since 2024
	• Toluwani Aremu (ML)	since 2023

MSc Students

- Juan Nicolas Sepulveda Arias (CV) since 2025
- Mohamed Mahmoud Mohamed Hendy (ML) since 2025
- Majid Mohamed Mohamed Sulaiman Ibrahim (ML) since 2025
- Prince Jha (ML) *[secondary]* (w/ *Salem Lahlou*) since 2025
- Muhammad Mohideen (CV) (w/ *Samuel Horvath*) since 2025

Alumni

- Tameem Bakr (ML) MSc · Mar 2026
- Shaikha Jasem Mohamed Ismail Alhosani (ML) MSc · Mar 2026
- Maryam Abdulla Rashed Abdulla Alshamsi (ML) MSc · Mar 2026
- Maryam Alshehyari (ML) MSc · Mar 2026
- Abdalla Khalid Mohamed Abdalla Alzaabi (ML) MSc · Mar 2026
- Ali Al-Ali (ML) MSc · Mar 2026